

Achraf Bitar

AI & Machine Learning Engineer | LLM systems and production data pipelines
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Lyon, France · 6-month final-year internship, March to September 2027 · Student visa granted, covering both coursework and the internship

PROFILE

Engineering degree from ENSMR, now finishing an MSc in Data Science at Centrale Lyon. I spent four months at ATOZ Services building a GDPR pseudonymisation service and the semantic search that runs on top of it, after a final-year project on the Attijariwafa Bank trading floor working on real-time financial flows. I write both Python and Java, and I take things all the way to deployment: I ran an experimental trading bot on AWS for over a year. Strong maths background (French CPGE, MPSI/MP). I'm looking for an internship where the model actually has to reach production.

PROFESSIONAL EXPERIENCE

ATOZ Services — AI Developer Mar – Jun 2026
ATOZ Group, tax and financial advisory (Luxembourg) · Casablanca, Morocco

- Built an end-to-end pseudonymisation service for financial documents: PDF text extraction (Poppler), OCR (Tesseract), MiniLM-L6 embeddings, storage and indexing in PostgreSQL/pgvector.
- Hybrid PII detection pipeline: GLiNER for named-entity recognition, regex rules for fixed formats, fuzzy matching for name spelling variants. Tuned to favour recall, since a missed entity costs far more than a false positive.
- Removed a synchronous LLM verification layer after measuring it: **it multiplied latency by 145 with no improvement to detection KPIs**. Moved to an offline check instead, at equal quality.
- Switched to an HNSW index on pgvector, which made semantic search fast enough to use interactively across the whole corpus.
- Internal Spring Boot / Angular application, including a chatbot that guides operators towards GDPR-compliant use of the service.
- Benchmarked NLP and embedding models to settle architecture trade-offs. Agile, Jira / Bitbucket / Confluence.

Attijariwafa Bank — Software Engineering Intern (final-year project) Feb – Jun 2025
Trading floor · Casablanca, Morocco

- Spring Boot micro-service for real-time monitoring of critical financial flows, pushed over WebSocket at **under 100 ms latency**.
- Ingestion and processing of high-frequency transaction logs; containerised deployment with Docker.

Vinci Energies (Exprom) — Junior Digitalisation Consultant Jun – Sep 2024
Rabat, Morocco

- Automated business processes on Power Platform with SQL Server integration. Removing the manual hand-offs cut processing time by **30%**.

PERSONAL PROJECT

Algorithmic gold trading bot (XAUUSD) Experimental project · Python AsyncIO · AWS EC2 · GPT-4o-mini · MetaTrader 5

- Personal project I designed, deployed and ran on my own on AWS EC2 for over a year, now retired. It read XAUUSD across H4/H1/M15 through GPT-4o-mini (macro regime, order blocks) and drove four external APIs concurrently — Finnhub, OpenAI, MetaTrader 5, Telegram — with dynamic position sizing.
- Over the period it ran: 45% win rate at a 1:3 risk-reward ratio, i.e. an expectancy of +0.8R per trade. Encouraging, but on too short a sample to draw a conclusion from. Alert-only architecture, no automated execution.

EDUCATION

École Centrale de Lyon — MSc Computer Science, Data Science track 2026 – 2027
ENSMR, Rabat — Engineering degree (Dipl. d'ingénieur d'État), Computer Science & Digital Engineering 2022 – 2025
CPGE, Meknès — MPSI / MP, intensive mathematics and physics 2020 – 2022

TECHNICAL SKILLS

AI & NLP: OpenAI API, Claude API, GLiNER, Presidio, MiniLM-L6 embeddings, RAG, semantic search, prompt engineering, model benchmarking
Languages & backend: Python (FastAPI, AsyncIO, Pandas, NumPy), Java (Spring Boot), SQL, REST APIs, Angular, React
Data: PostgreSQL / pgvector, HNSW indexing, Oracle, SQL Server, Redis
Cloud & tooling: AWS (EC2, S3), Docker, Git / GitLab CI, Linux, n8n, Power Platform, Jira / Confluence
Compliance: GDPR, pseudonymisation, data minimisation, technical documentation

LANGUAGES & CERTIFICATIONS

Arabic (native) · French (C1, TCF 549/699) · English (fluent, TOEIC 955/990)
IBM Python for Data Science. AWS Solutions Architect Associate in progress.